

CHAPTER 3

RESEARCH DESIGN AND METHODOLOGY

This section details the methodology employed in the development and evaluation of ForenXAI, an Explainable AI-driven framework with Retrieval-Augmented Generation for network log analysis. It describes the research design, the environment in which the study was conducted, the procedures for gathering and preparing data, the system architecture of the proposed tool, the tools and technologies used in its implementation, the step-by-step experimental procedures followed during development, and the evaluation metrics and validation techniques applied to assess the system's performance.

3.1 RESEARCH DESIGN

General Guidelines: Discuss thoroughly the research methodology pursued by the proponents. A research methodology describes the techniques and procedures used to identify and analyze information regarding a specific research topic. It is a process by which researchers design their study so that they can achieve their objectives using the selected research instruments. There are three types of research methodology—quantitative, qualitative, and mixed-method, which can be chosen based on the research objectives.

(add GenAI , indicate, RAG based)

This study follows a developmental research design, which is appropriate for studies whose primary objective is the creation, implementation, and evaluation of a functional technological artifact. Unlike purely descriptive or correlational research, developmental research involves the systematic process of designing, building, and testing a system against defined performance criteria. ForenXAI is the primary deliverable of this study, a working software tool rather than a theoretical model, and its success is measured by how well it performs against quantifiable evaluation metrics.

The study is also applied in nature, as it directly addresses a practical problem identified in both the literature and practice: the inability of current network forensic tools to automatically analyze PCAP/PCAPng-based network logs and produce transparent, human-readable explanations of detected threats. Rather than contributing solely to theoretical knowledge, this study produces a tool that forensic investigators can directly utilize in their post-incident analysis workflows.

A quantitative evaluation approach is used to measure the performance of ForenXAI's machine learning component, using standard classification metrics including accuracy, precision, recall, and F1-score, alongside confusion matrix analysis and out-of-distribution (OOD) generalization testing to evaluate the model's reliability beyond its training distribution. The usability and effectiveness of ForenXAI prototype are evaluated by cybersecurity experts and forensic analysts using the ISO/IEC25010 software quality framework, specifically across the Functional Suitability, Usability, Reliability, and Performance Efficiency characteristics. The interpretability of the XAI-generated explanations is assessed through structured expert evaluation, wherein the clarity, logical consistency, and usefulness of the explanations are rated against defined criteria. This mixed measurement approach ensures that both the technical performance of the detection system and the practical utility of its explanations are rigorously assessed.

3.2 RESEARCH ENVIRONMENT

General Description: The research environment establishes the technical and infrastructural context in which this study is conducted. It encompasses the datasets, computational resources, and devices employed to develop, evaluate, and validate the machine learning model and/or prototype development. Sub-sections are but not limited to the following:

- *Devices and Tools*
- *Datasets*

The research environment establishes the technical and infrastructural context in which this study is conducted, encompassing the datasets, computational resources, and devices employed to develop, evaluate, and validate ForenXAI's machine learning model and desktop prototype.

Devices and Tools

Table 2 reports the implementation environment used to develop, train, and evaluate ForenXAI's classification model.

TABLE 2a. Implementation Environment

Component	Specification
Python Version	3.11.9

Operating System	Windows 10 (Build 26200)
Processor	AMD64 Family 25 Model 124, AuthenticAMD
CPU Logical Cores	12
System RAM	15.3 GB
Development Environment	Visual Studio Code
Random Seed	42 (fixed throughout, for reproducibility)

TABLE 2b. Software Tools

Software Tools	Usage
Python	Base runtime
pandas, NumPy	Data manipulation and numerical operations
PyArrow	Parquet I/O and schema inspection
scikit-learn	Random Forest, Extra Trees, Decision Tree, MLP, Logistic Regression, SGD, stacking, evaluation metrics
LightGBM, XGBoost	Gradient boosting comparison models
Boruta	Feature selection
Optuna	Hyperparameter optimization
SHAP	Post-hoc explanation via TreeExplainer
CICFlowMeter v4	PCAP to flow-level feature extraction
Visual Studio Code	Development environment

Datasets

The training data for the machine learning model originates from two publicly available benchmark datasets selected to provide complementary coverage of network attack patterns. The CSE-CIC-IDS2018 dataset was developed through a collaborative project between the Communications Security Establishment (CSE) and the Canadian Institute for Cybersecurity (CIC). It contains 16,233,002 network-traffic instances collected over ten days in an AWS-based network environment and covers seven attack scenarios, including brute-force, Heartbleed, botnet, DoS, DDoS, web attacks, and network infiltration. The dataset also includes 80 traffic features extracted from captured network traffic using CICFlowMeter [SHARAFALDIN, LASHKARI, GHORBANI], [CANADIAN INSTI]. The TII-SSRC-23 dataset, introduced

by Herzalla et al. [19], contains 27.5 GB of PCAP-based network traffic comprising 32 traffic subtypes and 26 distinct attacks organized under four categories: DoS, brute-force, botnet, and information gathering.

The two datasets were merged because neither independently provides adequate coverage for the classification task. TII-SSRC-23 contains only 1,301 benign flows against 8,655,466 malicious records, a proportion insufficient for a classifier to learn a reliable representation of normal traffic, while CSE-CIC-IDS2018 supplies 13,390,249 benign flows. Conversely, CSE-CIC-IDS2018 contributes no reconnaissance class and no IoT botnet traffic, both of which TII-SSRC-23 provides. The merge is therefore complementary rather than merely additive. Table 3 summarizes the combined class distribution after merging.

TABLE 3. Combined Class Distribution

Class	CSE-CIC-IDS2018	TII-SSRC-23	Combined	Share
Benign	13,390,249	1,301	13,391,550	54.01%
DoS	1,918,233	7,490,929	9,409,162	37.95%
Info-Gathering	0	1,038,363	1,038,363	4.19%
Bruteforce	380,943	35,172	416,115	1.68%
Botnet	286,191	91,002	377,193	1.52%
Infiltration	160,639	0	160,639	0.65%
Web Attack	928	0	928	0.004%
Total	16,137,183	8,656,767	24,793,950	100.00%

This study curates and processes records from both datasets to isolate the target attack categories relevant to ForenXAI's scope: Denial-of-Service (DoS), brute-force attacks (FTP and SSH), botnet activity, web-based attacks, and network infiltration, as represented in CSE-CIC-IDS2018; and DoS, brute-force, botnet, and information-gathering attacks, as represented in TII-SSRC-23. Raw attack labels from both datasets were mapped to a unified seven-class taxonomy: Benign, Botnet, Bruteforce, DoS, Infiltration, Info-Gathering, and Web Attack.

Feature schema alignment between the two datasets was verified empirically rather than assumed. CSE-CIC-IDS2018 was extracted using CICFlowMeter V3,

which applies abbreviated column naming, while TII-SSRC-23 uses the V4 naming convention. A mapping of 48 column names reconciled the two, after which both sources shared exactly 78 feature columns with none unique to either. One discrepancy identified during this process was substantive rather than cosmetic: CSE-CIC-IDS2018 labels a field "CWE Flag Count," where the corresponding TCP flag is CWR (Congestion Window Reduced); TII-SSRC-23 uses the correct designation, consistent with prior work documenting systematic faults originating within the CICFlowMeter tool itself.

3.3 DATA GATHERING PROCEDURES

3.3.1 Data Collection Strategy

General Guidelines: Describe who the participants are, how they were chosen, and why they were selected in which different groups of participants may be involved for various research-related activities (interview, pre-surveys, system testing, system usability evaluation, system expert's validation or system acceptance). State the sampling method (e.g., purposive sampling, convenience sampling). Identify the participants where the number is typically between 5–15 for each participant category (e.g., owner, employee or staff) as this range provides valuable insights without excessive redundancy. Finally, provide a narrative that justifies the sample size, explains why the chosen participants are representative, and identify any inclusion or exclusion criteria.

3.3.2 Data Interpretation and Analysis Strategy

General Guidelines: This section discusses how the collected data will be processed and interpreted from various research activities (interview, surveys, usability evaluation, expert's validation). An example is the use of the following:

§ Likert-scale data analysis involves quantifying user responses to identify overall strengths and weaknesses. Responses are numerically coded (e.g., 1–5), and descriptive statistics such as mean scores per question or per category are computed. Aggregated results highlight which aspects of the system users found most usable and where improvements are needed.

§ Interpreting the results requires using standard statistical measures to determine central tendencies and variability in responses.

3.3.3 Ethical Considerations

General Guidelines: This section explains how the data from various research activities (interview, surveys, usability evaluation, expert's validation) will be securely stored, maintained, and protected. It covers both digital and physical data, ensuring

compliance with ethical research standards. An example is Data management practices include anonymizing participant information, securing survey responses in encrypted files or password-protected systems, and controlling access to data only for authorized researchers.

The narrative should describe how data will be organized (e.g., coding schemes for qualitative data, spreadsheets for quantitative data), how long it will be retained, and when it will be disposed of. Transparency in data management ensures research integrity, participant trust, and compliance with institutional or legal requirements.

3.4 SYSTEM ARCHITECTURE / PROPOSED MODEL

General Guidelines: In this section, illustrate (provide a diagram) and discuss the technical aspect of the prototype and/or model using System Architecture (e.g., client-server, three-tier, or layered).

3.5 EXPERIMENTAL PROCEDURES

General Description: The Experimental Procedures section of a thesis manuscript explains the systematic process undertaken to design, develop, implement, test, and evaluate the proposed Machine Learning (ML) model and/or prototype system. This section must provide sufficient technical and methodological detail to allow other researchers to replicate the study, validate the findings, and assess the reliability of the experimental results. The discussion should be written in a formal academic style using the past tense because the procedures have already been conducted. Sub-sections are but not limited to the following sections:

- *Experimental Setup (Model Training and Testing Environment, Prototype Development and Deployment Environment)*
- *Data Pre-Processing Procedures (if applicable)*
- *Model Training and Testing Procedures (if applicable)*
- *Model Performance Evaluation Procedures (if applicable)*
- *Model and Prototype Integration Procedures (if applicable)*

